ECHO: The Signal That Was Never Sent

Entangled Minds: The Quantathon — Case Report

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Mandatory Modules: 01, 03 and 05
Stretch Module: 06 — The Impossible Claim

1. Final Theory and Verdict

My verdict is that the ECHO experiments do not demonstrate faster-than-light communication. The results show that quantum circuits can produce predictable measurement outcomes, strong correlations between entangled qubits, and successful quantum-state teleportation when the required correction information is available.

My confidence is high in the scientific conclusion that entanglement alone cannot transmit a controllable message faster than light. However, the notebook does not establish what happened to Dr. Zara Mahmud or identify anyone responsible for her disappearance. Those remain open investigative questions.

2. Key Evidence and Clues

Clue 1 — Predictable quantum operations (Module 01). The X gate produced the expected outcome of 1, while applying H twice returned the qubit to its initial state, producing 0. A single H gate produced approximately balanced outcomes: 537 zeros and 463 ones over 1,000 shots. The three-qubit circuit also produced the target result 101 in all 1,000 shots.

Clue 2 — Strong quantum correlations (Module 03). The Bell-pair circuit produced 1,015 outcomes of 00 and 985 of 11 in the Z basis. In the X basis, it produced 1,007 outcomes of 00 and 993 of 11. The additional CHSH test reported (S=2.84), exceeding the local classical bound of 2 under the standard CHSH assumptions. This supports the presence of correlations inconsistent with local-hidden-variable models under the test's assumptions, but it does not demonstrate faster-than-light communication.

Clue 3 — Message recovery and teleportation (Modules 05 and 06). The majority-voting decoder recovered the intended message, ECHO. In the teleportation test, receiver verification produced 2,000 zeros with correction operations, whereas the circuit without corrections produced 1,007 zeros and 993 ones. These results illustrate the importance of the correction operations for the tested input state.

3. Evidence Board

Category	Finding
FACT	Module 01 produced the expected deterministic outcomes and approximately balanced superposition measurements.
FACT	Module 03 reported Bell correlations and a CHSH value of (S=2.84).
FACT	Module 05 decoded the intended message as ECHO.
FACT	Module 06 produced different receiver statistics with and without correction operations.
CLAIM	Zara believed ECHO could transmit information faster than light.
INFERENCE	The observed correlations and teleportation results do not establish a faster-than-light communication channel.
OPEN QUESTION	Who configured the final experiment, and what happened to Zara? The circuit results cannot resolve these questions.

4. Experiments and Results

Module 01 — The Gatekeeper. I tested X, H, H–H and a three-qubit circuit targeting 101. The deterministic tests produced 1, 0 and 101 as expected. The H-gate test produced 537 zeros and 463 ones.

What it proves: Quantum gates can produce deterministic outcomes or probabilistic measurement distributions depending on the prepared state.

Module 03 — The Two Observers. I prepared a Bell pair and measured correlations in the Z and X bases. The recorded results showed strong correlations, and the CHSH calculation returned (S=2.84).

What it proves: The measured correlations support a quantum explanation and, if the CHSH implementation and measurement assumptions are valid, challenge local-hidden-variable explanations. Correlation is not equivalent to communication.

Module 05 — The Damaged Message. I grouped the supplied 96 bits into 32 triples and applied majority voting. The decoder recovered ECHO.

What it proves: A three-bit repetition code can correct a single bit flip in each triple, but it can fail if two or more bits in a triple are corrupted.

Module 06 — The Impossible Claim (Stretch). I compared teleportation with and without the receiver's correction operations. The corrected circuit passed the reported receiver verification, producing 2,000 zeros; the circuit without corrections produced approximately balanced outcomes.

What it proves: For the tested input and circuit, the correction operations are necessary to obtain the intended receiver result. The experiment illustrates why shared entanglement alone is insufficient for completing the teleportation protocol.

5. Interpretation and Alternative Explanation

The combined evidence supports the conclusion that ECHO's proposed faster-than-light communication mechanism has not been demonstrated. Module 03 investigates quantum correlations, while Module 06 examines the role of classical correction information in teleportation. Neither experiment establishes that a controllable message can travel faster than light.

A plausible alternative explanation is that Zara performed a valid quantum experiment but misinterpreted correlated outcomes as evidence of faster-than-light signalling. This interpretation is more consistent with standard quantum mechanics than the claim that entanglement alone can transmit a usable message.

Nevertheless, the scientific results cannot establish why Zara disappeared. Experimental error, incorrect interpretation, altered records or deliberate intervention remain possibilities rather than proven explanations.

6. Next Experiment and Limitations

My next experiment would independently validate the CHSH circuit, verify its measurement settings and outcome mappings, and calculate the four required correlation terms from the raw counts. I would repeat the measurements with sufficient shots to estimate statistical uncertainty. A valid, statistically significant CHSH violation would challenge local-hidden-variable explanations but would still not demonstrate faster-than-light communication.

I would also test teleportation using several input states and compare the receiver's measurement statistics with and without the classical correction information.

A key limitation is that simulator outputs represent a modeled circuit and do not independently verify events in a physical laboratory. Finite measurement counts introduce statistical uncertainty. The teleportation experiment uses a particular input state, so its results should not be presented as a complete experimental demonstration for every unknown quantum state. The notebook also cannot identify a suspect or establish the circumstances of Zara's disappearance.

7. References and AI Assistance

1. *ECHO: The Signal That Was Never Sent*, Entangled Minds: The Quantathon, participant manual, supplied by the organizers.
2. *Code of Conduct — Quantathon*, supplied by the organizers.
3. IBM Quantum Learning, quantum computing and quantum teleportation: https://learning.quantum.ibm.com/

AI assistance: OpenAI ChatGPT was used to help review the notebook's logic, clarify quantum-computing concepts, and organize and edit this report. The reported numerical results were taken from the supplied execution summary and should be cross-checked against the saved notebook outputs before submission.

Final conclusion: The reported experiments support quantum correlations and the role of classical information in teleportation, but they do not establish faster-than-light communication. The scientific verdict is distinct from the unresolved questions surrounding Zara's disappearance.